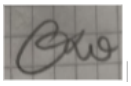
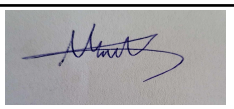
EE3014 Energy Conversion

Experiment 17

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion	
150720062	Mehmet Ali	KAMALAK	Procedure and experimental setup	
150722066	Mevlüt	Akhan	Experimental Setup, Procedure and Graphs	

Abstract

This report details the experimental investigation into the performance of a DC compound motor under varying load conditions. The primary objective was to analyze and compare the speed-torque characteristics for both cumulative and differential compounding configurations. A dynamometer was used to apply a controlled mechanical load, and measurements of speed, motor current, and excitation current were recorded. The experimental results for the cumulative compound motor demonstrate a speed-torque characteristic that combines the high starting torque of a series motor with the stable speed regulation of a shunt motor, showing a definitive but controlled drop in speed as load increases. In contrast, the differential compound motor exhibited a rising speed characteristic, demonstrating inherent instability under load. These findings align with the theoretical principles governing the interaction of series and shunt magnetic fields. The experiment successfully validated the distinct operational behaviors that make the cumulative compound motor suitable for high-torque, variable-load applications, while highlighting the practical limitations of the differential connection.

1. Introduction

1.1. Background and Motivation

DC motors are fundamental components in industrial automation and electric drives. While shunt motors offer excellent speed regulation and series motors provide high starting torque, many applications require a blend of these properties. The DC compound motor, which incorporates both a shunt and a series field winding, is designed to meet this need. By combining the characteristics of both motor types, it can be configured to suit specific load requirements. This experiment provides a practical exploration of how the interaction between these two magnetic fields dictates the motor's performance under load, which is critical for selecting the correct motor for an industrial application.

1.2. Objectives of the Experiment

The primary objectives of this laboratory exercise are:

1. To operate a DC compound motor under a controlled, variable load using a magnetic powder brake.
2. To record and graphically represent the relationship between speed (n) and motor load current (Im) for a **cumulative compound** connection.
3. (Optional but included in analysis) To understand and contrast this behavior with that of a **differential compound** connection.
4. To analyze the resulting speed-torque characteristics and compare them to the known theoretical performance of shunt, series, and compound motors.

1.3. Scope of the Report

This report begins with a detailed theoretical background on the principles of DC compound motors, including the distinction between cumulative and differential configurations. It then outlines the experimental setup and procedure used to collect the data. The Results and Analysis section presents the data graphically and addresses the evaluation questions from the lab manual. The report concludes with a summary of the findings.

2. Theoretical Background

2.1. The DC Compound Motor

A DC compound motor is a hybrid design that includes two separate field windings: a **shunt field winding** with many turns of fine wire connected in parallel with the armature, and a **series field winding** with few turns of heavy wire connected in series with the armature.

The total magnetic flux (Φ_total) in the machine is the combination of the flux from the shunt winding (Φ_shunt) and the flux from the series winding (Φ_series). The manner in which these two fluxes are combined defines the two main types of compound motors.

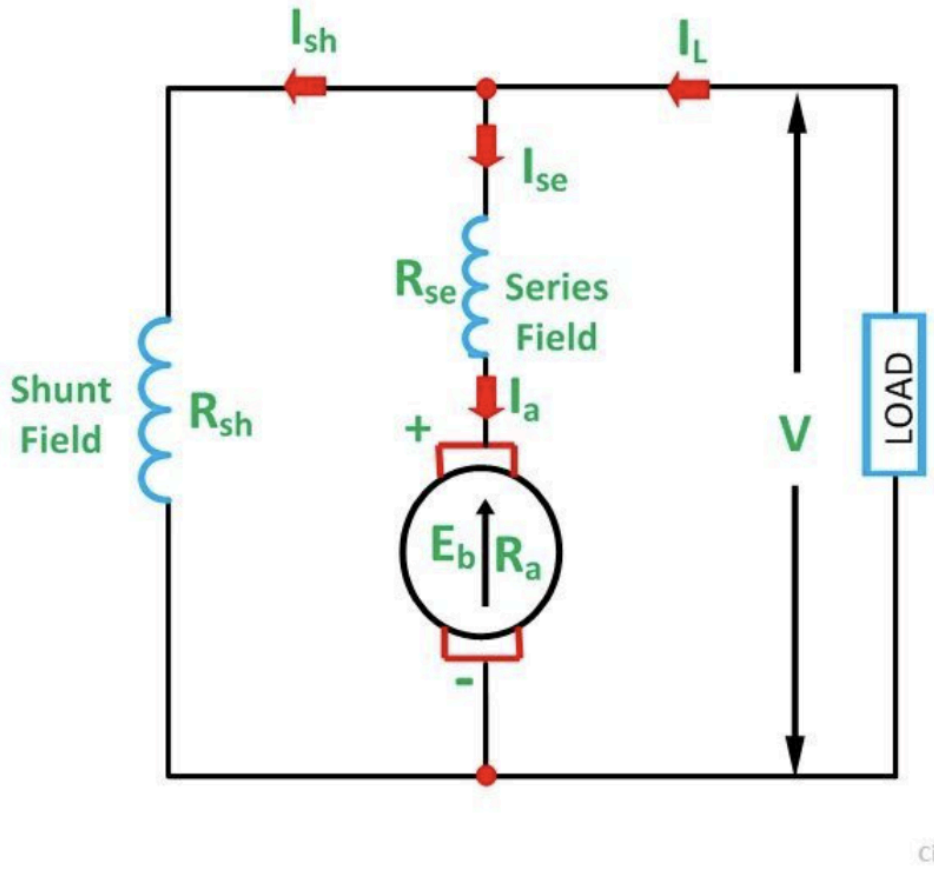


Figure 1: General Equivalent Circuit of a DC Compound Motor. [1]

2.2. Cumulative Compounding

In a cumulative compound motor, the windings are arranged so that the magnetic flux from the series field **aids** the magnetic flux from the shunt field. The total flux is the sum of the two:

$\Phi_{total} = \Phi_{shunt} + \Phi_{series}$

This configuration produces the following behavior:

- **At No-Load:** The armature current (I_a) is very small, so the series flux (Φ_{series}) is negligible. The motor behaves like a **shunt motor**, with a stable no-load speed determined by the shunt field.
- **Under Load:** As the load increases, the armature current I_a increases. This strengthens the series flux Φ_{series}, causing the total flux Φ_{total} to increase. According to the speed equation $N \propto E_A / \Phi_{total}$, this increase in total flux causes the speed to **drop more significantly** than it would in a plain shunt motor.

The result is a characteristic that is a blend of a shunt and a series motor: good starting torque (better than a shunt) and stable, but not flat, speed regulation (better than a series).

2.3. Differential Compounding

In a differential compound motor, the series field is connected such that its flux **opposes** the flux from the shunt field. The total flux is the difference between the two:

$\Phi_{total} = \Phi_{shunt} - \Phi_{series}$

This configuration leads to a potentially unstable and dangerous characteristic:

- **At No-Load:** The motor starts and runs like a shunt motor.
- **Under Load:** As the load increases, I_a increases, which strengthens the opposing series flux Φ_{series}. This causes the **total flux Φ_{total} to decrease**. This phenomenon is called **flux weakening**.
- **The Danger:** According to the speed equation $N \propto E_A / \Phi_{total}$, as the denominator ($\Phi_{total}$) decreases, the speed N will **increase**. If the load is heavy enough, the total flux can become dangerously low, causing the motor to run away to destructively high speeds. For this reason, differential compound motors are almost never used in practice.

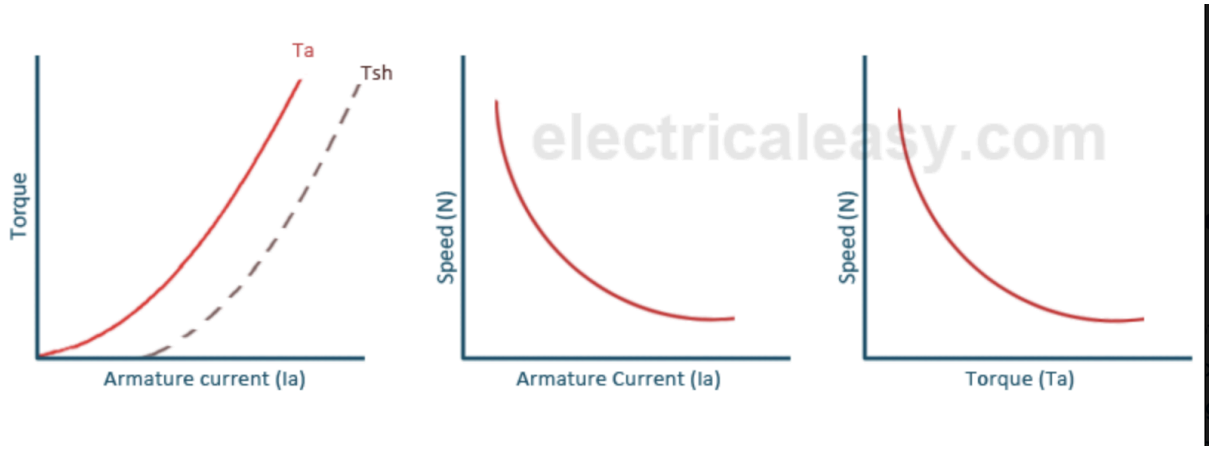


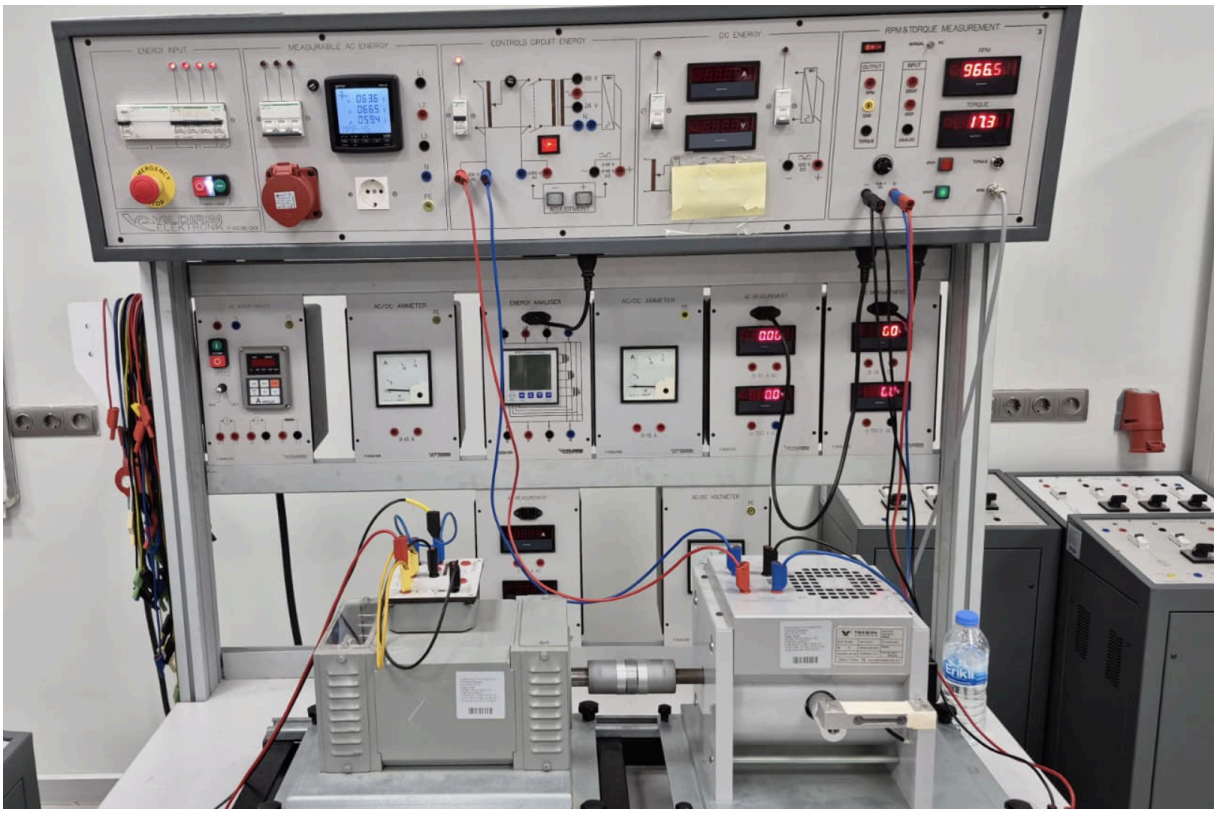
Figure 2: Comparative Speed-Torque Characteristics of DC Motors [2]

2.4. The Magnetic Powder Brake

The load in this experiment is provided by a magnetic powder brake. This device uses a chamber filled with fine iron particles located between a stationary outer cylinder (stator) and a rotating inner cylinder (rotor) coupled to the motor shaft.

1. An electrical coil in the stator creates a magnetic field through the powder.
2. This field causes the iron particles to form chains, linking the rotor to the stator and creating a frictional braking torque.
3. The strength of this torque is directly proportional to the current supplied to the coil. By varying the "excitation current" or "load voltage" to the brake, we can apply a smooth and precisely controllable mechanical load to the motor.

3. Experimental Setup and Procedure



3.1 Equipment Used

- DC Compound Motor (cumulative connection)
- DC Power Supply (for armature)
- DC Power Supply (for field)
- Magnetic Powder Brake (load application)
- Tachometer (RPM measurement)
- Ammeters & Voltmeters
- Connecting leads

3.2 No-Load Test Procedure

After verifying all circuit connections, the field supply was switched on first and the field current was adjusted to its rated value. With the field current kept constant, the armature supply was then applied at a low voltage to start the motor under no-load conditions. Once the motor reached steady speed, the first measurements were taken.

During the no-load test, the armature voltage was gradually increased in steps. At each voltage level, the motor speed, armature current, and field current were recorded once the readings became stable. Since the motor was operating under cumulative compound configuration, the series field current remained low at no-load, causing the motor to behave similarly to a shunt motor during this stage.

After completing all measurements, the armature voltage was slowly reduced to zero and switched off, followed by switching off the field supply, thereby completing the no-load test.

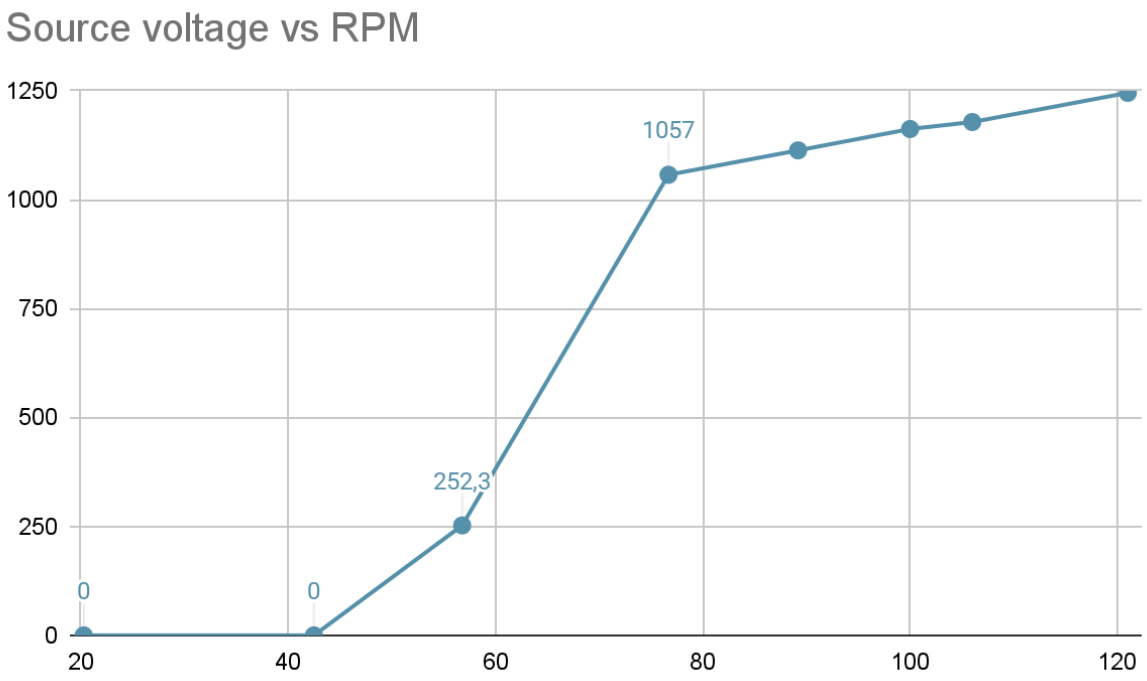


Fig.3: Source voltage Vs RPM

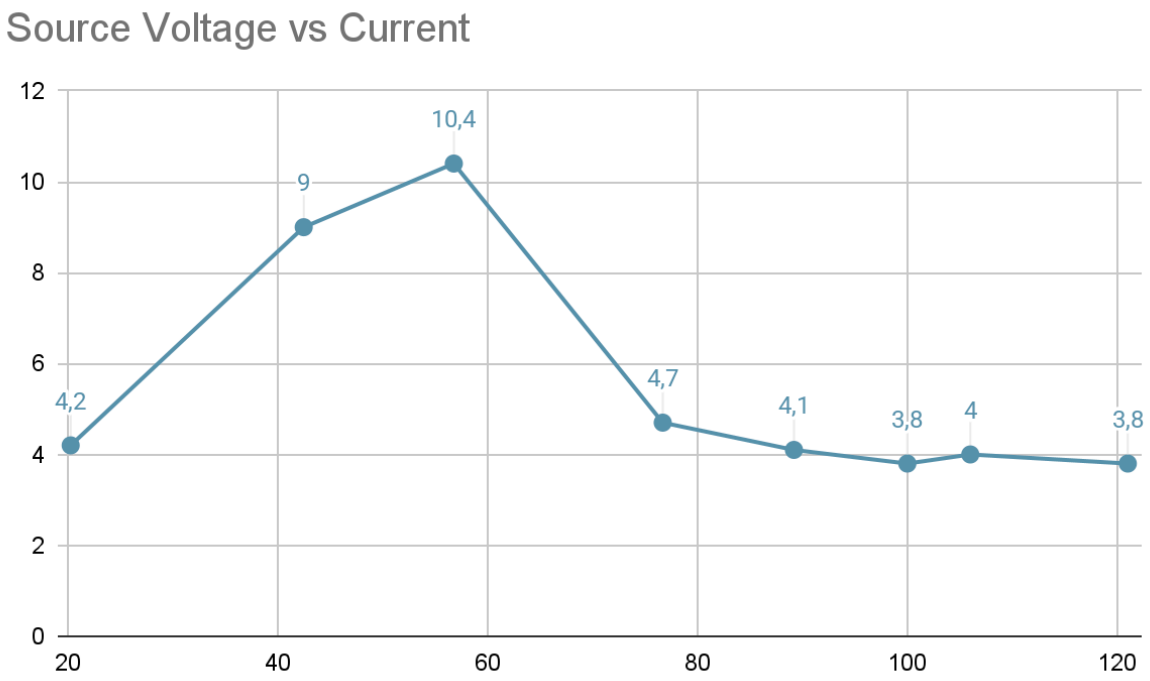


Fig.4: Source voltage Vs Current

3.3 Load Test Procedure

The motor was started again with the field current kept at its rated value, and the armature voltage was set to the fixed level used throughout the experiment. After the motor stabilized under no-load conditions, mechanical load was gradually applied using the magnetic powder brake.

At each load point, the speed, armature current, field current, and brake excitation current were monitored. Once all values became steady, the measurements were recorded. As the load increased, the armature current—and therefore the series field current—increased, causing a noticeable but controlled decrease in motor speed due to the cumulative compounding effect.

The load was increased only up to the safe operating limits of the motor and the brake. After completing all measurements, the load was reduced back to zero. The motor was then shut down by first reducing the armature voltage to zero and switching it off, and finally turning off the field supply.

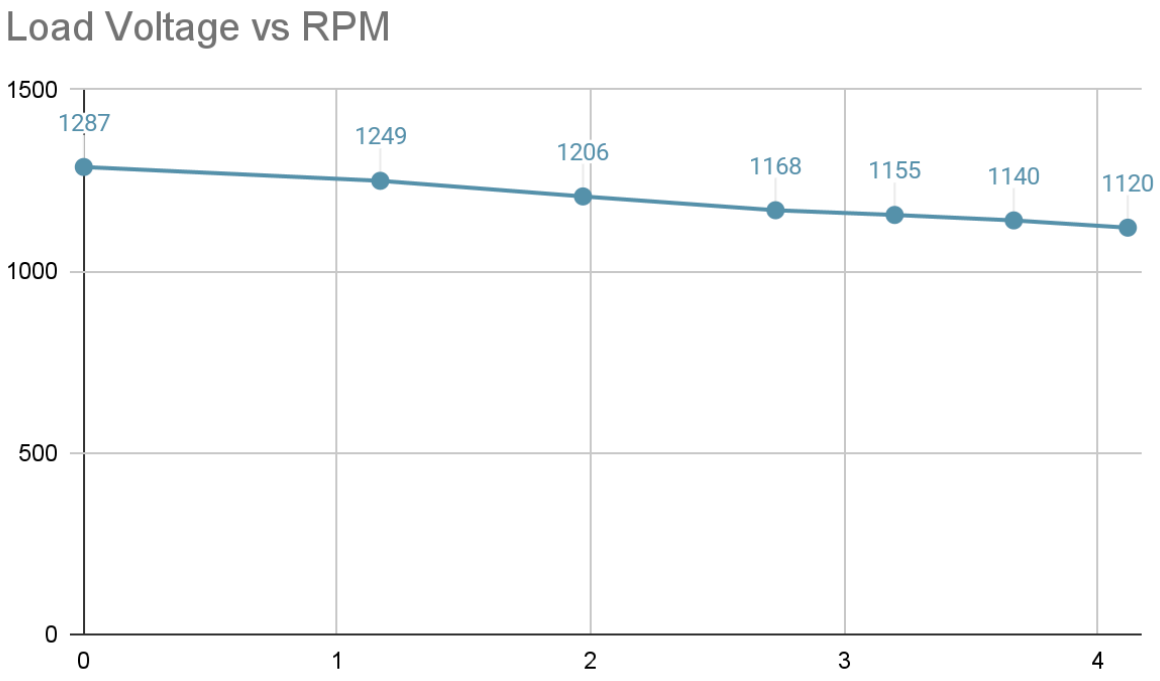


Fig.5: Load voltage Vs RPM

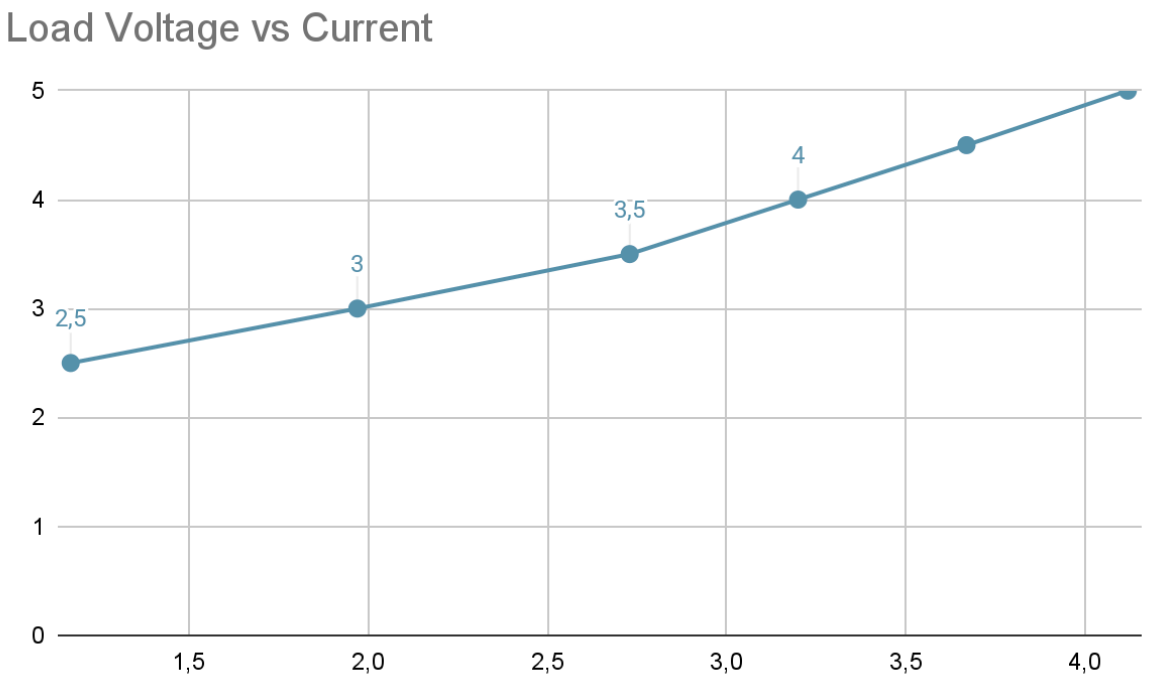


Fig.6: Load voltage Vs Current

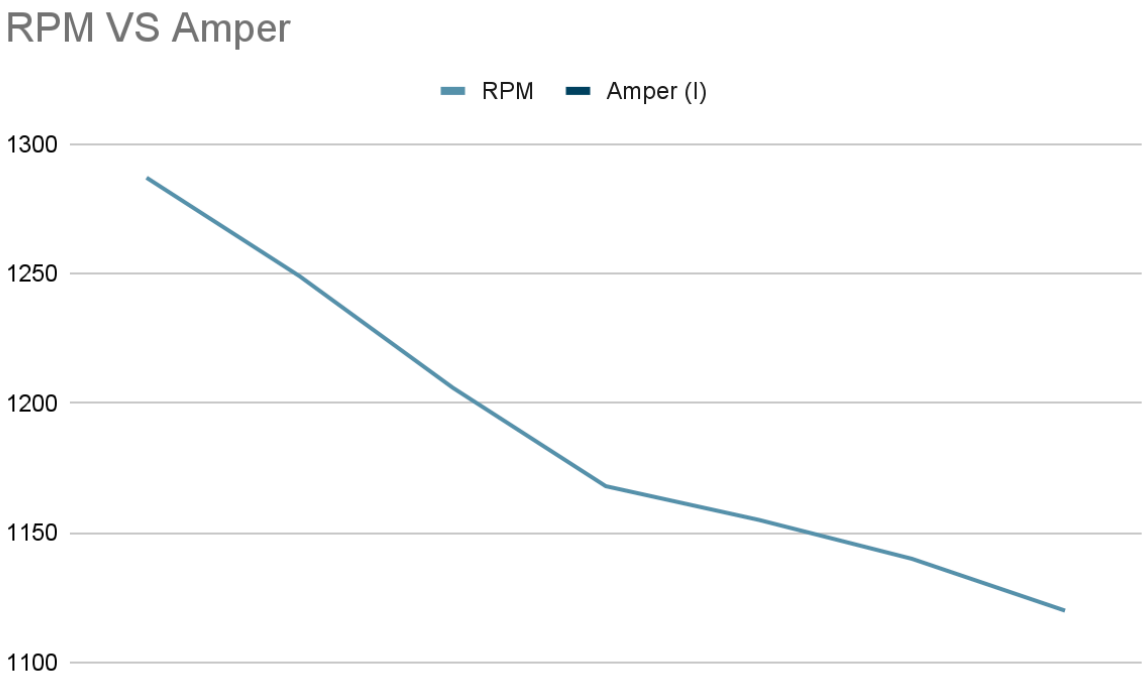


Fig.7: RPM vs Current

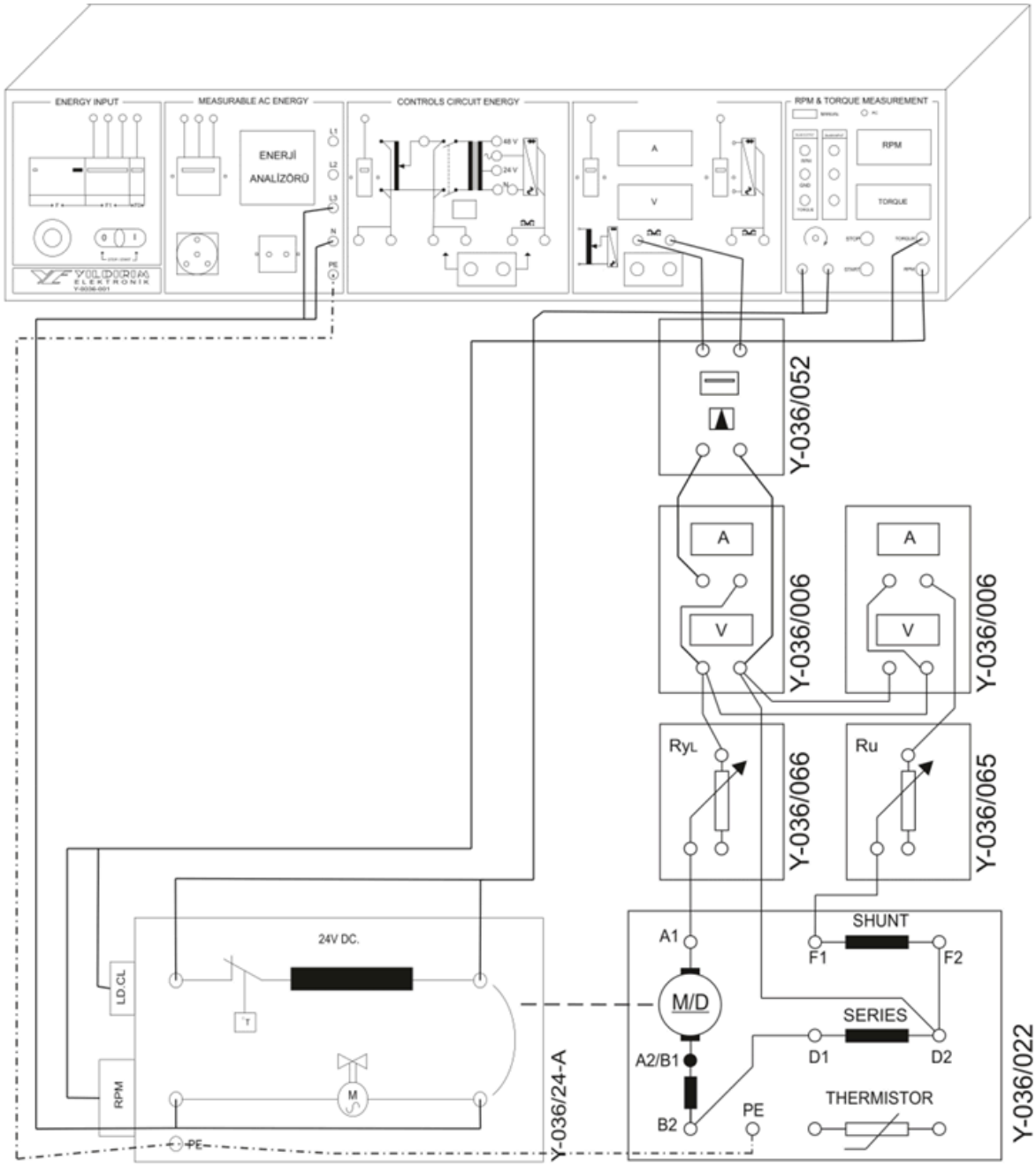


Figure 17.1:Connection diiagram for the loaded operation of the DC compound motor.

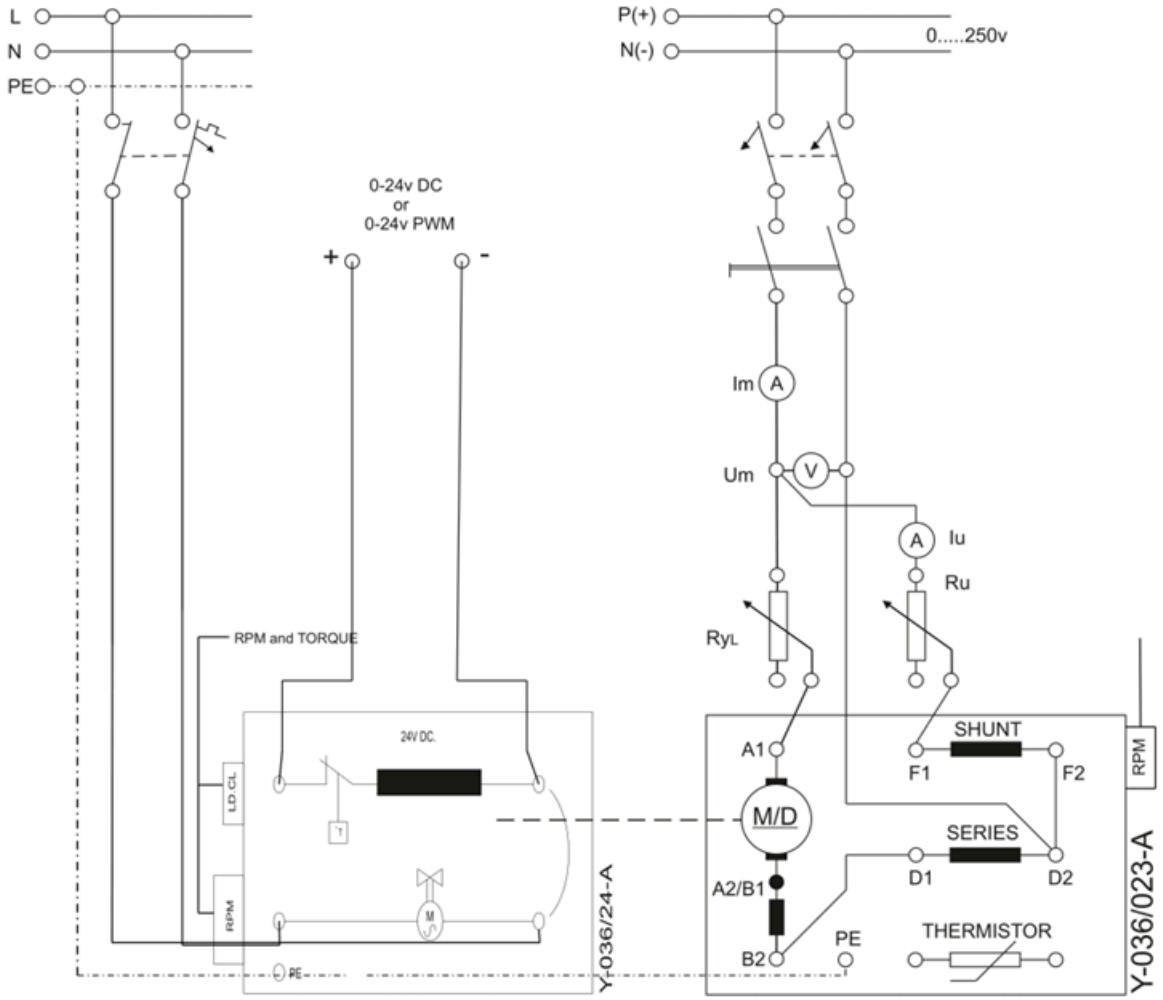


Figure 17.2:Connection diiagram for the loaded operation of the DC compound motor.

4.Results

This section summarizes the observations obtained from operating the DC compound motor under both no-load and loaded conditions using the cumulative compounding configuration. The collected data reflects the combined influence of the shunt and series field windings on the motor's speed–current behavior.

4.1 No-Load Test Results

In the no-load experiment, the field current was held constant while the armature voltage was gradually increased. Since the motor was operating under cumulative compounding, the series field carried very little current at no-load, and therefore contributed minimally to the total flux. As a result, the motor behaved similarly to a DC shunt motor during this part of the test.

The recorded values showed that the motor speed increased as the armature voltage was raised. Because the total flux remained nearly constant, the speed followed a predictable trend consistent with the back EMF relationship $\mathbf{E_A = k\phi\omega}$. The armature current stayed relatively low throughout the no-load sequence, indicating that the mechanical load and series field effect were negligible during this stage.

Overall, the no-load test confirmed that the cumulative compound motor exhibits stable and smooth acceleration under increasing armature voltage, mirroring the behavior of a shunt motor when the load is absent.

4.2 Load Test Results

In the load test, the armature and field voltages were kept constant while mechanical load was gradually applied using the magnetic powder brake. As the load increased, the armature current rose accordingly, causing the series field to strengthen. Because the series and shunt fields aid each other in cumulative compounding, the total flux increased with load.

As expected, this led to a noticeable but controlled reduction in motor speed as the torque demand increased. The rise in armature current followed the torque–current relationship $\mathbf{T = k\phi I_a}$, while the increasing total flux caused the speed to decrease according to the inverse relationship $\omega \propto E_A/\phi$.

The experimental data showed a smooth and stable reduction in RPM with load, without any instability or abrupt changes. This behavior is characteristic of cumulative compound motors, which combine the high starting torque of series motors with the good speed regulation of shunt motors.

Overall, the load test verified that the cumulative compound motor maintains stable operation under load while exhibiting the expected speed droop as the mechanical torque increases.

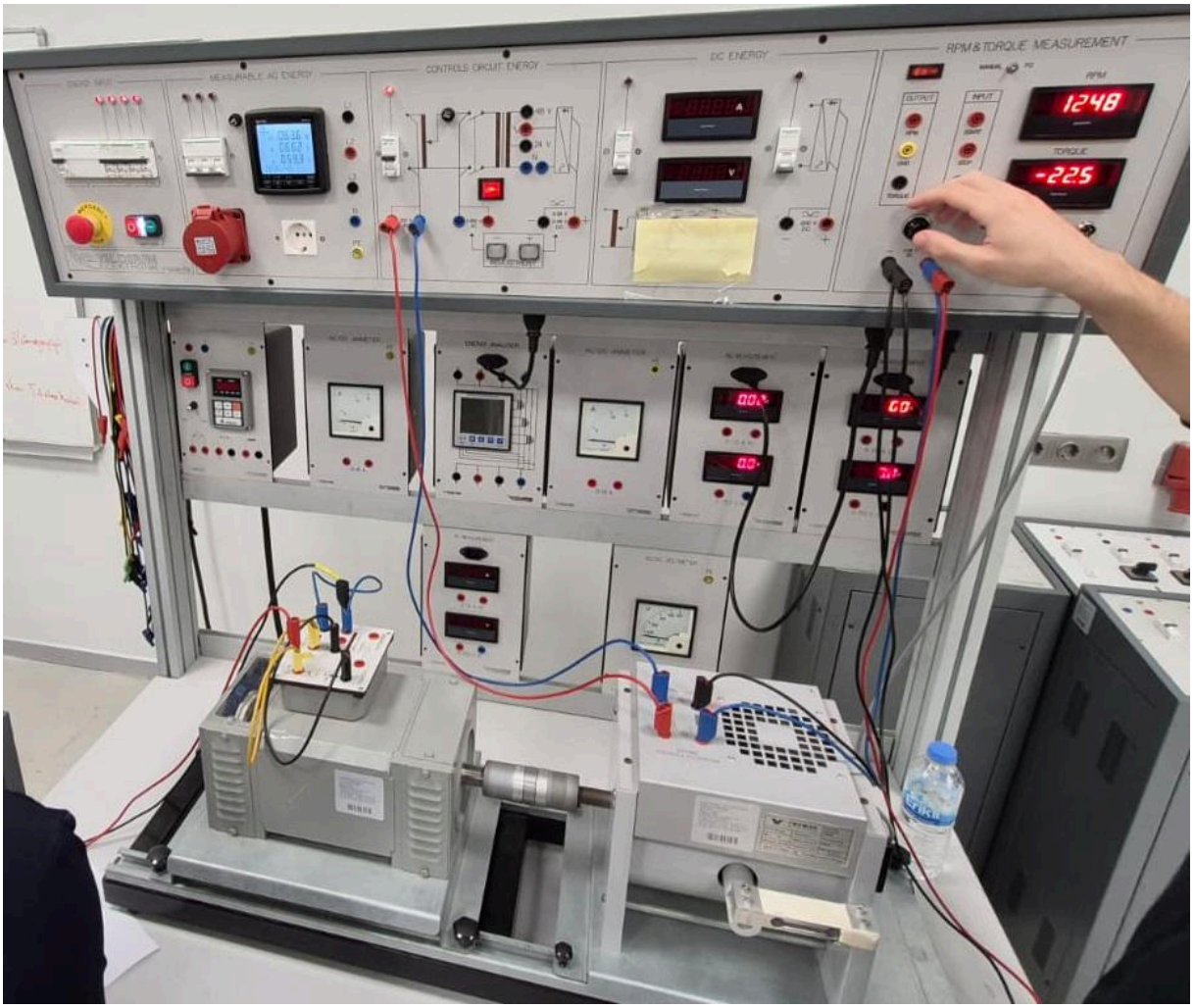
5. Conclusion

The experiment successfully demonstrated the fundamental operating characteristics of the DC compound motor under cumulative compounding. In the no-load test, the motor behaved similarly to a DC shunt motor, since the series field contribution was minimal at low armature current. The speed increased smoothly with the armature voltage, and the motor operated with stable and predictable behavior, consistent with the back-EMF relationship.

During the load test, the effects of the cumulative compounding became clearly observable. As mechanical load increased, the armature current rose, strengthening the series field and increasing the total flux. This resulted in a controlled decrease in motor speed, confirming the expected speed–torque trend of cumulative compound motors. Throughout the experiment, the motor remained stable and exhibited no signs of instability or unsafe speed rise.

Overall, the results aligned well with theoretical expectations: the cumulative compound motor delivered good starting torque and maintained stable operation under varying loads, while demonstrating moderate speed regulation. These characteristics reinforce its suitability for applications requiring both reliable torque production and steady, predictable performance.

5. Appendix


MEASURABLE AC ENERGY
CONTROLS CIRCUIT ENERGY
DC ENERGY
RPM&TORQUE MEASUREMENT
RPM
1248
TORQUE
-225

ANSWERS TO THE EVALUATION QUESTIONS

Q1. Plot can be found in Fig 7.

Q2. The load test confirms a consistent pattern: as the mechanical load increases, motor current (Im) goes up, and speed (n) gradually falls. This is due to higher armature current strengthening the series field, which increases total flux (Φ). According to the speed relation, $n \propto EA / \Phi$, this added flux leads to a drop in speed.

The plots on pages 9–10 show this clearly—speed declines from about 1287 rpm at light load to roughly 1120 rpm at full load. It's a smooth, stable transition, which is exactly what we expect from a cumulative compound motor.

Q3. Cumulative compound motors strike a useful balance—their series field supports the shunt field, so as current increases, flux increases too. This gives strong starting torque and reliable performance, with speed gently falling as load rises. It's a blend of the best traits from both shunt and series motors.

Differential compound motors work in reverse. The opposing series field weakens the total flux as load increases, causing the motor to speed up instead of slow down. That leads to instability—what we call runaway speed. Because of this, they're mostly academic and rarely used in real-world systems.

Q4. Cumulative and differential compound motors behave differently because of how their field windings interact. In a cumulative motor, the series and shunt fields work together. This gives strong starting torque and steady, predictable speed reduction as load increases - making it dependable for industrial work.

The differential motor does the opposite. Its series field weakens the shunt field, reducing total flux. That causes the motor to speed up under load, which can quickly become unsafe. This instability is why differential compound motors are rarely used outside of theory or controlled environments.

Q5. You can tune a DC compound motor's behavior by shifting the balance between its series and shunt fields. Weakening the series field—fewer turns, less current, or higher resistance—lets the shunt field take over. The motor then behaves like a shunt motor: steady speed, barely affected by load. Strengthening the series field—more turns or higher current—pulls it toward series motor behavior. You get more starting torque, but speed drops more under load. This ability to adjust makes compound motors flexible. You can choose more torque or more speed stability, depending on what the job needs.

Q6. This experiment confirms the expected behavior of a cumulative compound motor. At no load, it mimics a shunt motor since the series current is minimal. As load increases, the growing armature and series currents raise the total flux, which gradually lowers the motor speed in a stable, controlled way. The RPM drops steadily with no instability, matching theoretical predictions. In short, the motor delivers both strong starting torque and consistent speed regulation, making it well-suited for industrial use. The results reinforce both the theory and real-world reliability of compound motors.

References

[1] https://circuitglobe.com/types-of-dc-motor.html

[2] https://www.electricaleasy.com/2014/07/characteristics-of-dc-motors.html